

Classification algorithm in various situation

eg → KNN

Common?

$D_x \rightarrow (x, y)$ input → predict

Train → ? → Test

(KNN)

good x
Bad

← (D_x)
Train

simple

solve → $\sqrt{}$
← { imbalanced data }

credit-card

call →

Historical

Data

	income	# of loan	cs	Status
c1	—	—	—	N
c2	—	—	—	N
c3	—	—	—	//
...				
c10k	—	—	—	N

9000 → N ✓ (90%)
1000 → Yes ✓ (10%)

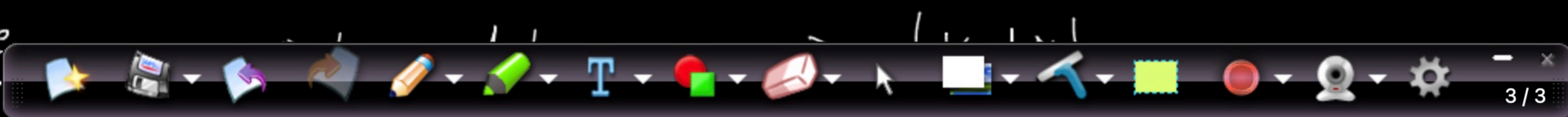
Weak learners

$D_X \rightarrow (X, Y)$
↑
Balanced
dataset
↑
only → $\boxed{1}$

$\hookrightarrow \in [0, 1]$
↑
50%
~~45%~~
20%

~~50%~~ 70%

imbalance



$D \rightarrow (x, y)$

normalise = True

df.y.value_counts()

0 → counts
1 → counts

imp?
 80%
 X

(-ve) 90%
 100
 +4
 10%

Random
 100

100 -ve
 100
 → Balanced

(ii) oversampling

interviews
 93%

90%

situation

(i) undersampling

90%

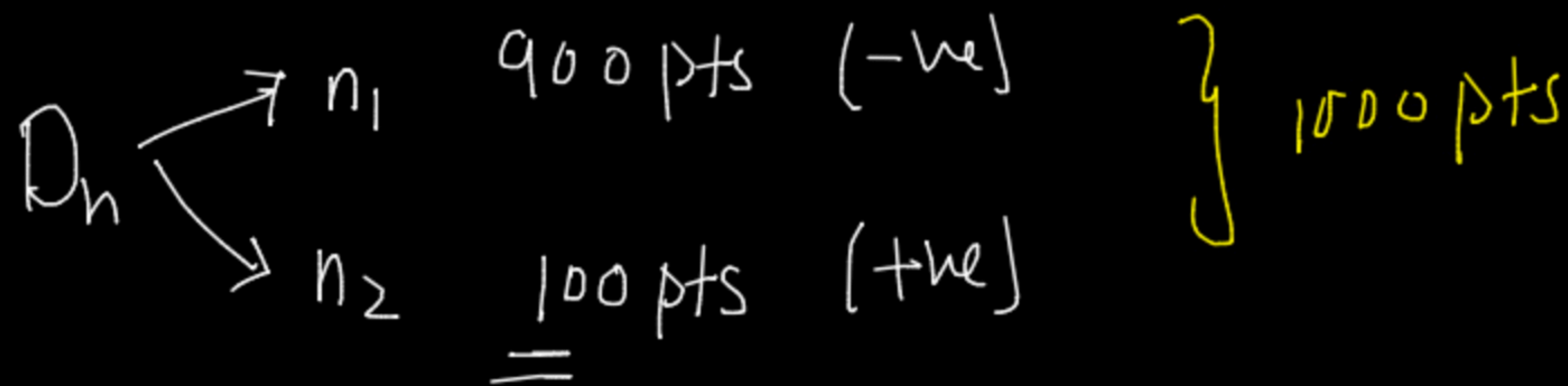


$D_n (x, y)$
 (n_1, n_2)

$n_1 \rightarrow 100 (+ve) \rightarrow$ minority
 $n_2 \rightarrow 900 (-ve) \rightarrow$ majority

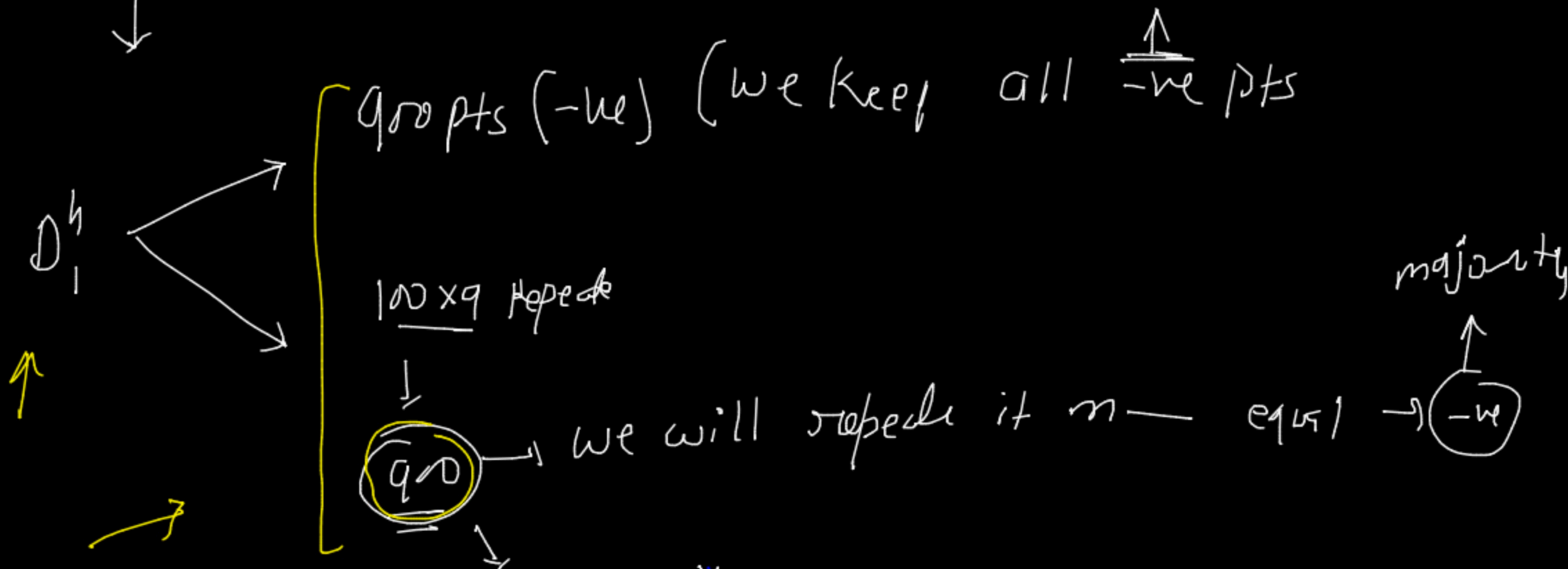
KNN(
f()

Oversampling



majority class

imbalanced



1800 pts → (900)

Sklearn $\rightarrow$ KNN

parameters



class-weight



100 +ve pts

900 -ve pts

Row $\rightarrow$ [_____]
[_____]
⋮
[_____]

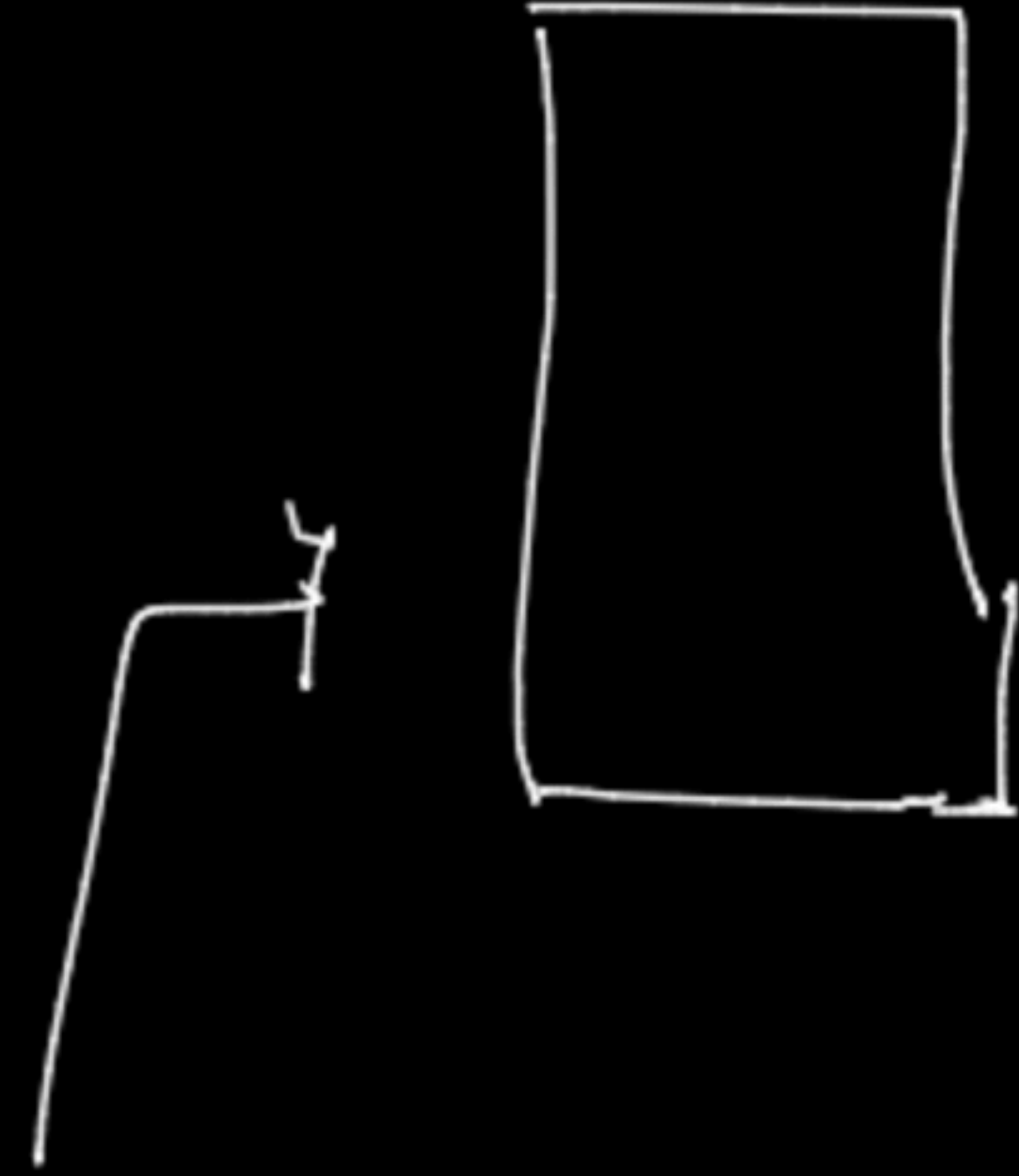
w_{+} $\rightarrow$ more weight to minority class

$w_{-} = 1$ less

1s
0s
 $\rightarrow$

$\underbrace{100 \text{ pts}}_{\uparrow \text{ testing}} \rightarrow \begin{cases} 95 \rightarrow -ve \rightarrow \underline{84} \% \\ 5 \rightarrow (+ve) \rightarrow X \end{cases} \rightarrow \begin{matrix} \swarrow \\ 84 \% \text{ score} \\ \uparrow \\ \underline{cool} \end{matrix}$

$\uparrow$
 X
 user



⇒ Multiclass classifier

↳ (i) Binary classification

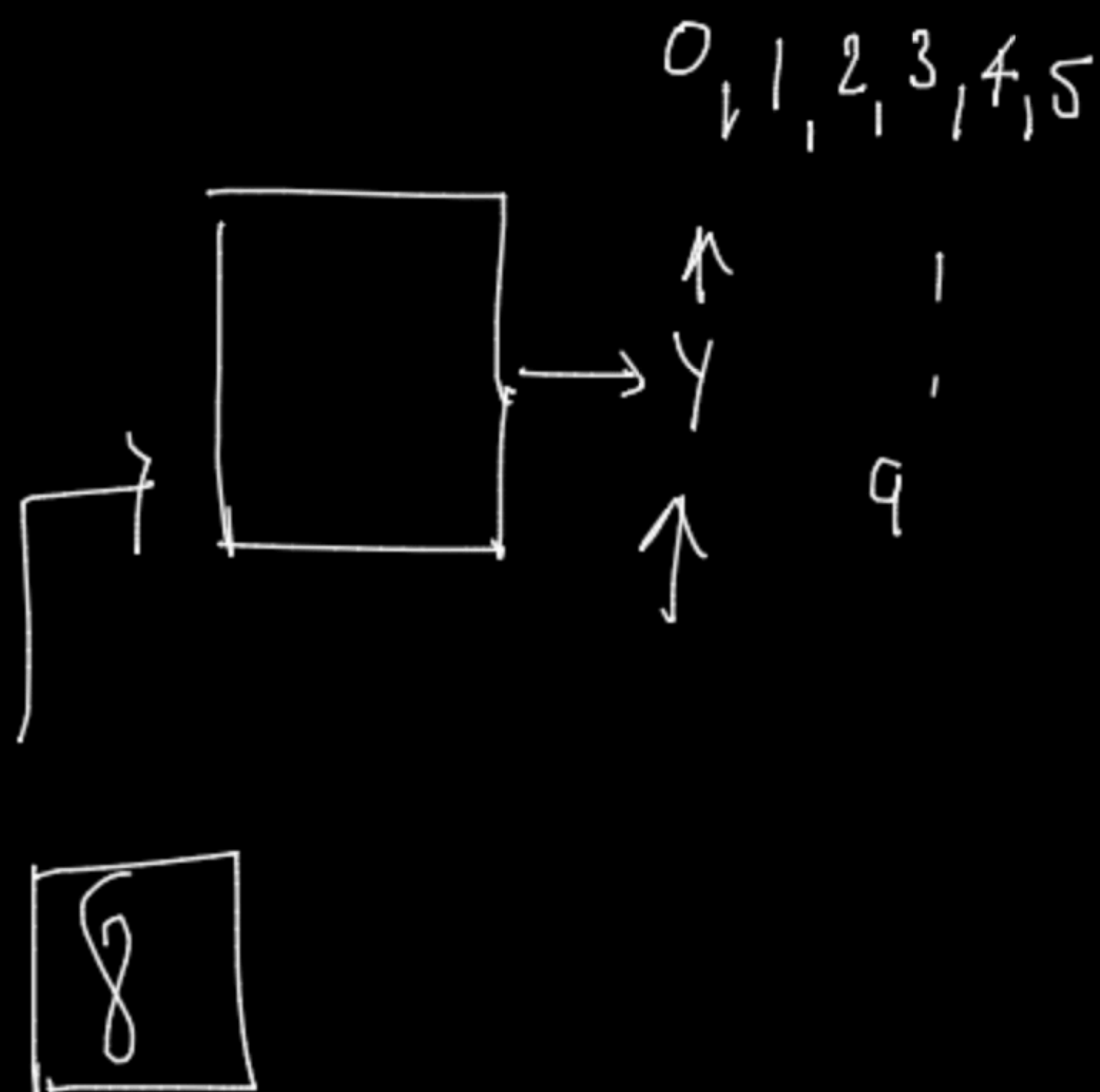
$$Y \in [0, 1]$$

(ii) Multiclass classification

$$Y \in [0, 1, 2, 3 \dots k]$$

Q. KNN

Yes



eg.

Hand-wr

X
0

9

2

3

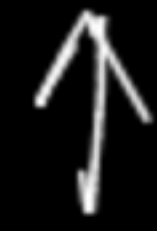
Y
0

1

2

3

$D(x, y)$



$\in [0, 1, 2]$



$KNN(n=7) \rightarrow$



predictions



majority

(7) pts

x_{query}



11

=

0	1	2
3	4	0
		1



mod len



=

train test split

↓

ii "Random"

time ↑

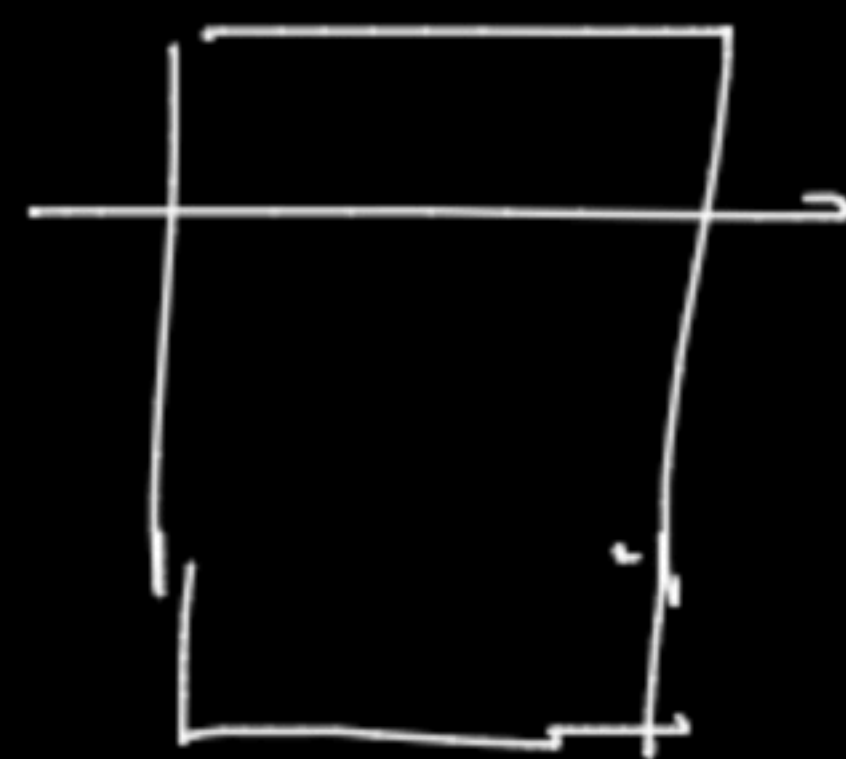
(iv) time-based splitting

imp

"stratified"

↑ classification

problem
↓
X
↑



0 → 70%
1 → 30%

→ KNN()

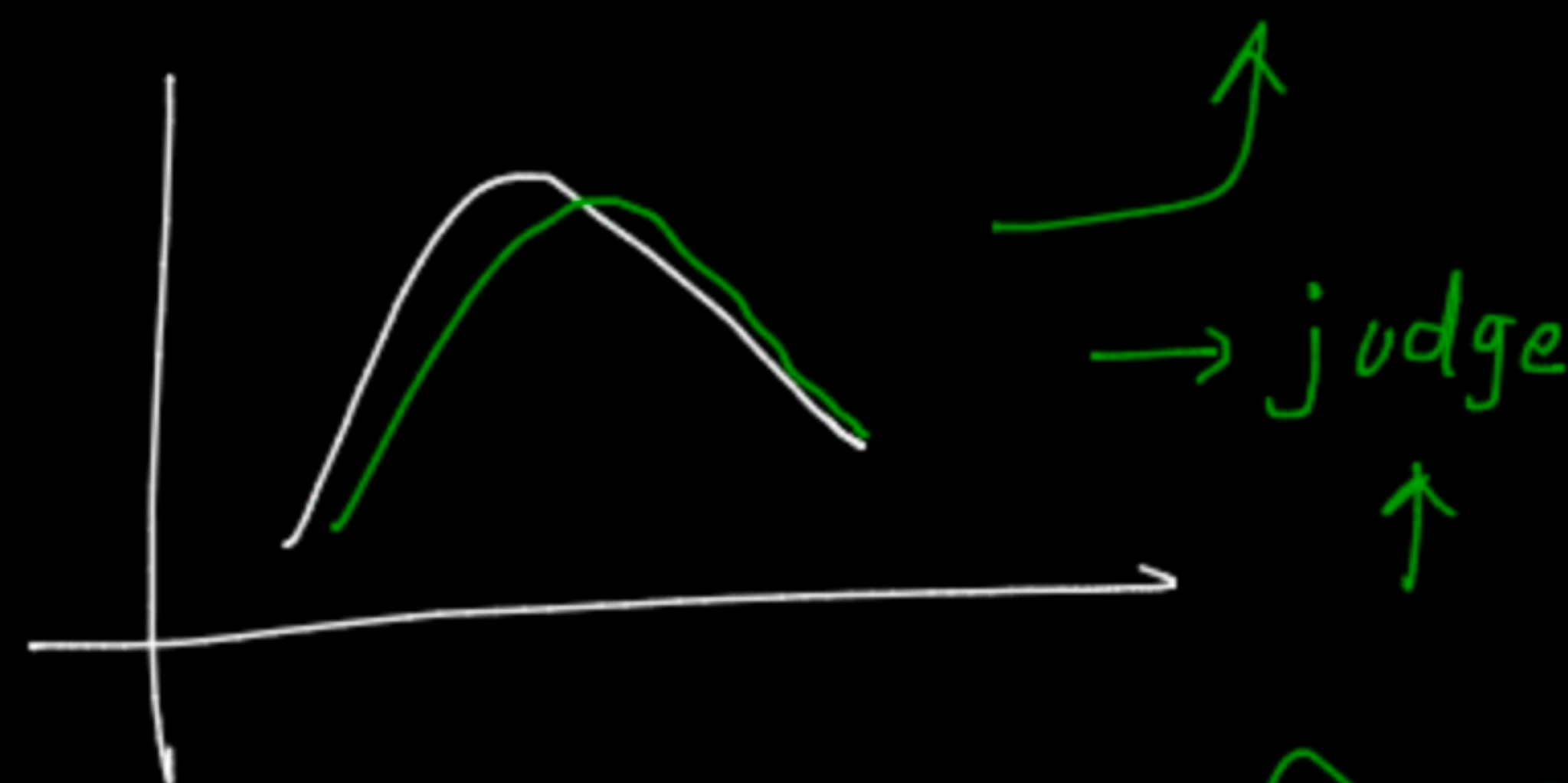


→ testing
0 → 40%
1 → 60%

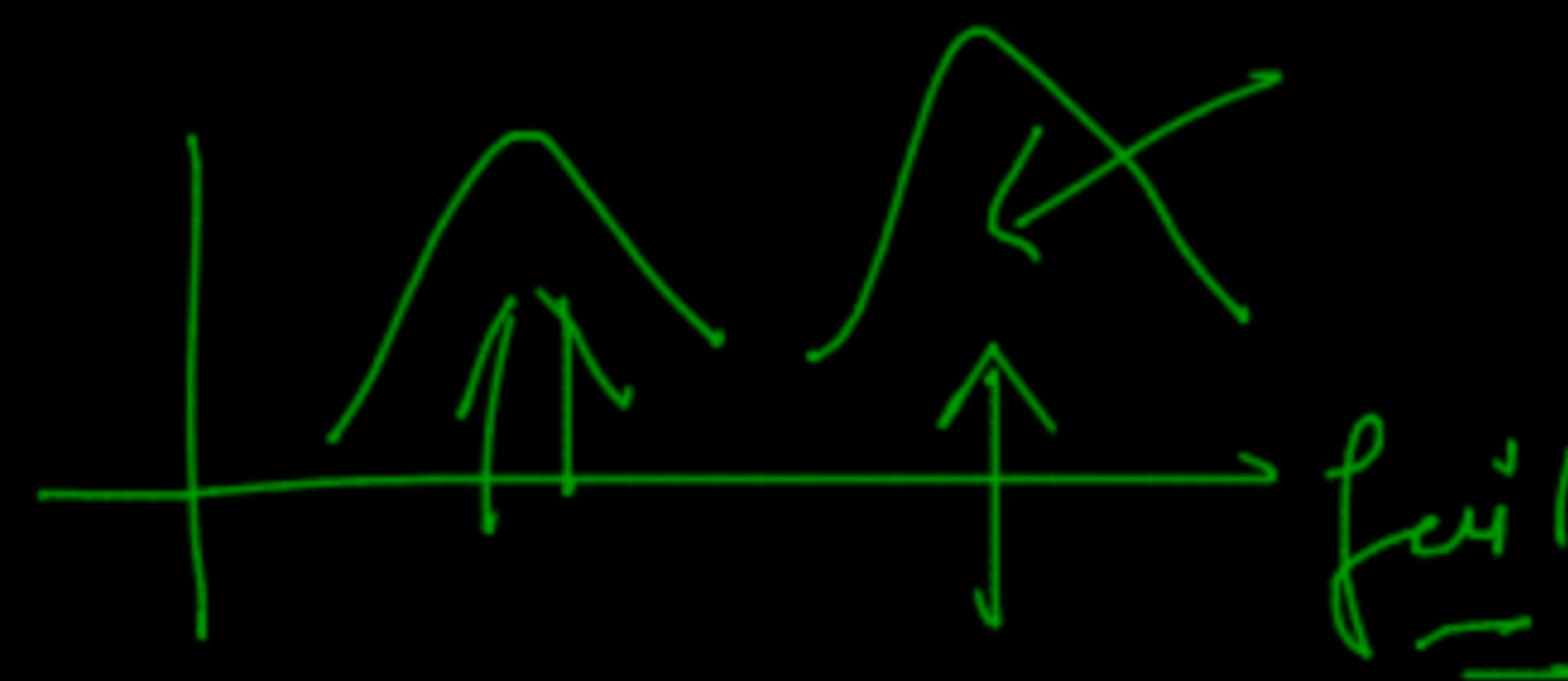
↑
Errors

↑
↑

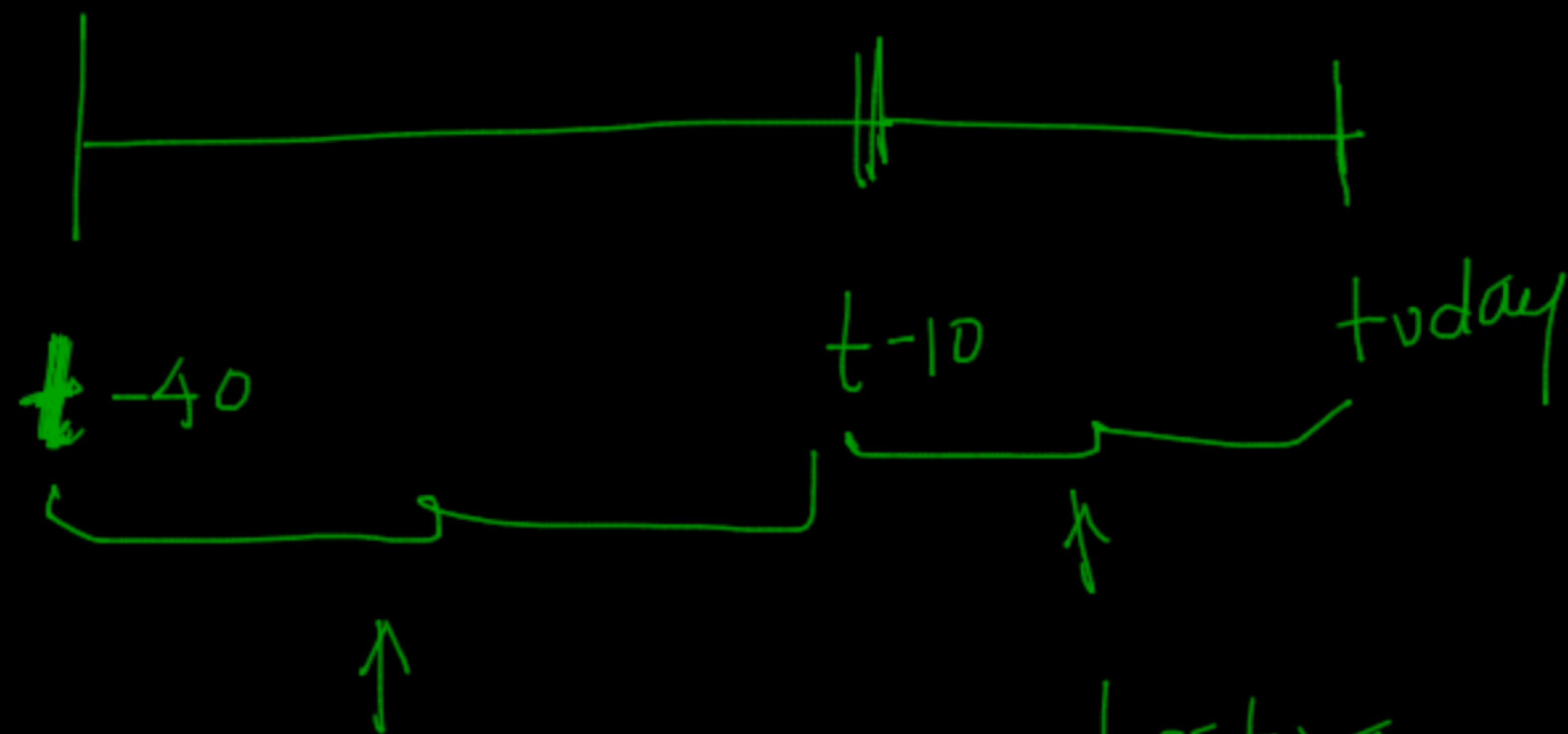
→ 90%
↑
cool
↑



→ judge
↑



Sal



model train

testing

date

3 month

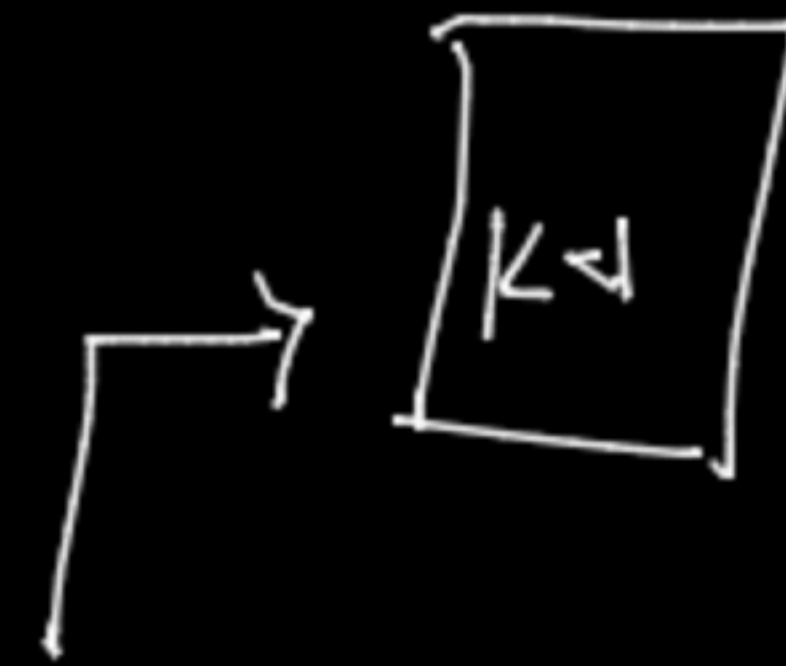
testing

impact of outliers

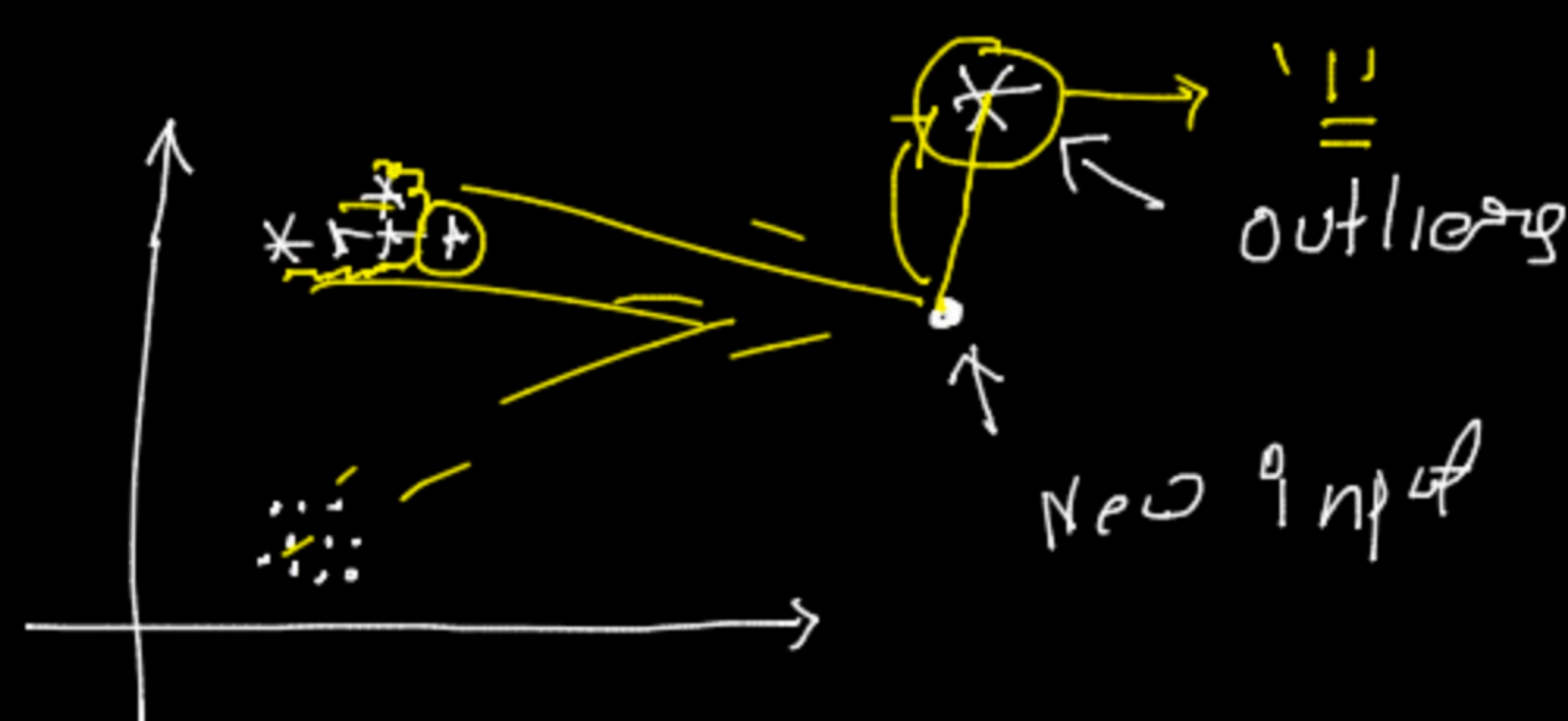
KN N $\rightarrow$ Distance $\xrightarrow{\text{eg.}}$

(30, +30)

stats $\rightarrow$ Box-plot
outlier detection



input $\rightarrow$ '1'



solution $\updownarrow$

5



story

$\downarrow$

Data

$\downarrow$

remove outlier

preprocess $\rightarrow$ standard-scales

Data

suggestions

model
↑
uniform
↑
standard

diff → 501
→ 80%
↑

But
↑

$x_3 (23, 1)$

	t_1	t_2
	(-)	(-)
	(-)	(-)
x_1	23	0.2
x_2	28	0.2
	23	1.0

(0-100)

(0-1)

effect → diff

$$\underline{\text{distance}}(\bar{x}_1, \bar{x}_2) \Rightarrow 5$$

$$\text{distance}(x_1, x_3) \Rightarrow 0.8$$

(0.8) → $x_1 \rightarrow x_3$
↓
?

figure?



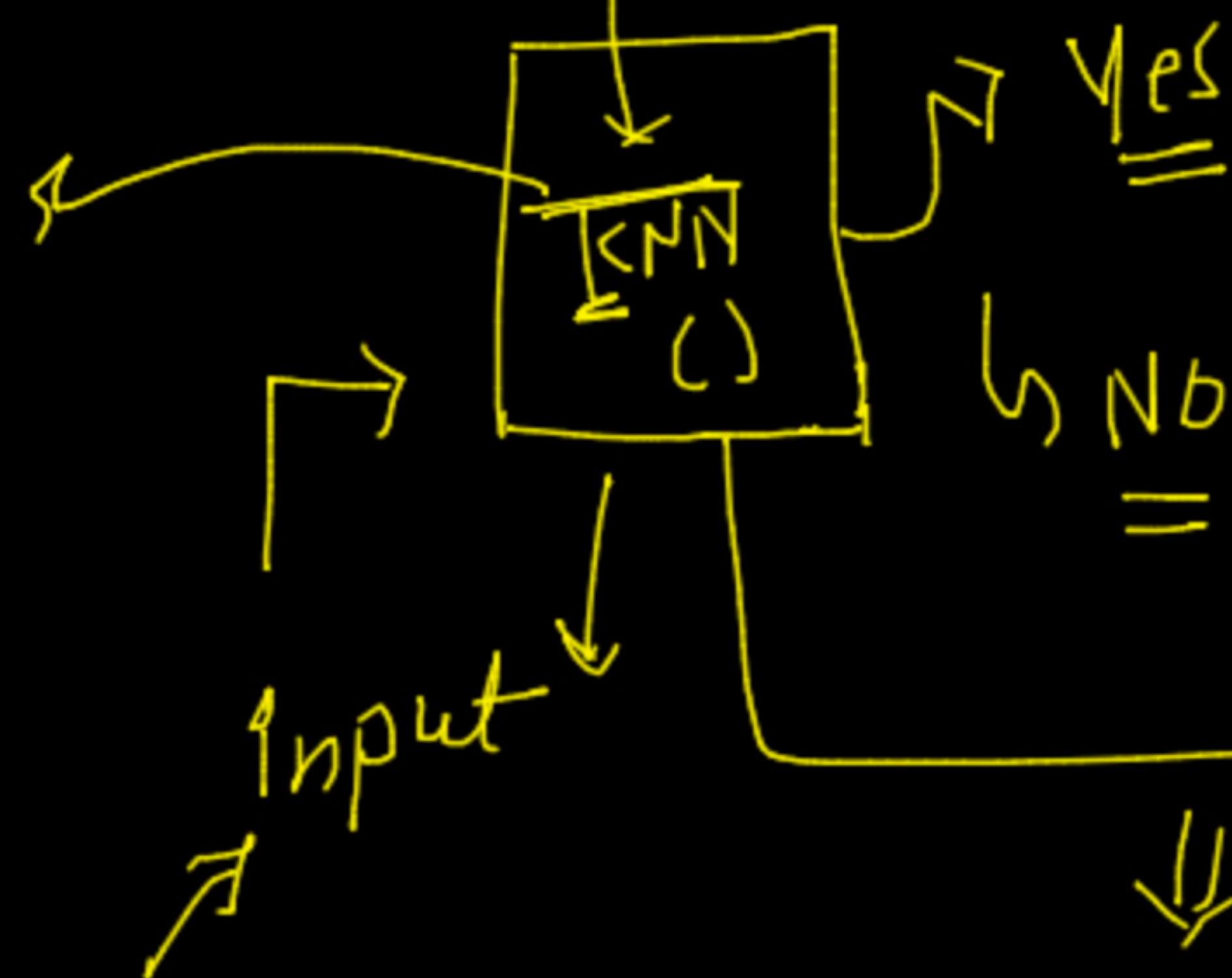
$(x_1) \rightarrow (23, 0.2)$
↑
 $(x_2) \rightarrow (28, 0.2)$

diff % → 0
diff → $\frac{28-23}{23} \Rightarrow 5\%$
↑

Banking
↑

$D_x(x, y)$
↑ ↑

histo



model interpretability
vs
blackBox

Reason
↑

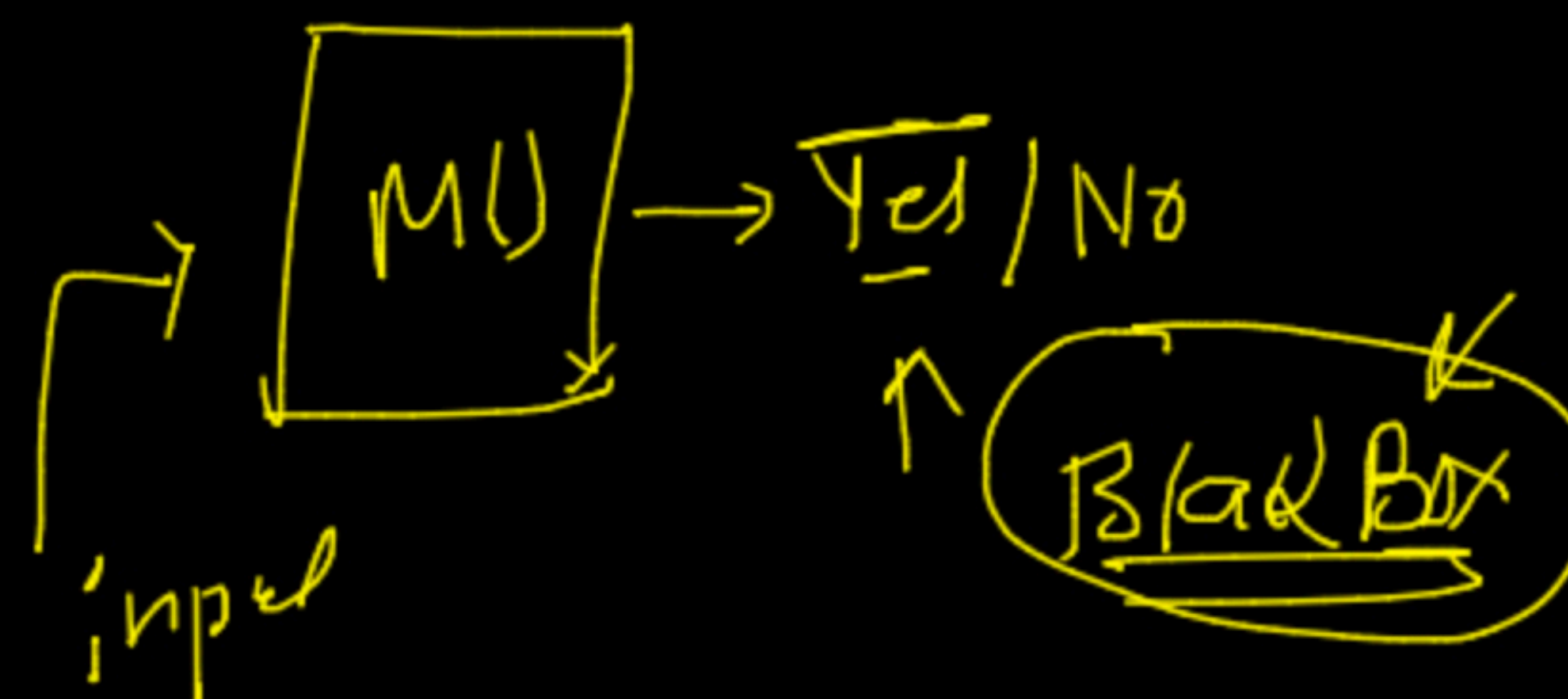
'NN'
↑

bcz

$\{ \begin{array}{l} CS < 650 \\ income < 20 \end{array} \}$
→

$\left\{ \begin{array}{l} CS \\ income \\ \# \text{ of active loan} \end{array} \right\}$

Blackbox



Low Battery
Your Mac will sleep soon unless plugged into a power outlet.

Feature importance → Forward selection → RFE →

↓ (MI) → (KNN)

$(f_1 - f_{100})$

↓
Usefull x

$(f_1, f_2, \dots, f_{10})$

64% 63%

D → (100 feature, Y)

Step 1

KNN

$f() \rightarrow Y\text{-pred}$

testing

test

↑ good

↑ good
(10) KNN()

EDA
↓
Remove

(x_1, y)
↑
 (x_2, y)

testing

KNN

$f'() \rightarrow 68\% \uparrow$

64% ↓

→ (10, Y)

50%

↓
(50 f, Y)

RFE

(x_1, y)
↑
 (f_1, f_2)

→ 54% →

ILP

$f()$

$(f_1, f_2, \dots, f_4)$